

Unit 2: Reading, Analyzing and Interpreting Maps

2.1 Space Categorization on a Map

Space categorization involves dividing a geographic area on a map into distinct zones or categories based on specific characteristics or criteria. This process helps to organize complex spatial data into meaningful groups, making it easier to analyze and interpret.

Examples of categorization criteria include:

- **Land use:** Residential, commercial, agricultural, industrial areas
- **Population density:** High, medium, low population zones
- **Vegetation type:** Forests, grasslands, wetlands

By categorizing space, GIS users can identify patterns, relationships, and trends within the mapped area, which supports better decision-making and planning.

2.2 Levels of Measurement

In GIS, data is classified into different levels of measurement that determine how it can be represented, analyzed, and symbolized:

- **Nominal:**
 - Data is categorized without any specific order.
 - Represents names or types.
 - **Example:** Land use types (residential, commercial), soil types.
- **Ordinal:**
 - Data categories have a meaningful order but the intervals between categories are not equal.
 - **Example:** Risk levels such as low, medium, and high.
- **Interval:**
 - Numeric data with equal intervals between values but no absolute zero point.
 - Allows addition and subtraction but not multiplication or division.
 - **Example:** Temperature measured in Celsius or Fahrenheit.
- **Ratio:**
 - Numeric data with a meaningful zero point, allowing all mathematical operations.
 - **Example:** Population count, distance, area size.

2.3 Relationship Between Symbolology and Data Measurement

Symbolology in GIS refers to the use of visual symbols—such as colors, shapes, and sizes—to represent spatial data on maps. The choice of symbols depends directly on the **level of data measurement**, ensuring that the map conveys information clearly and accurately.

- **Nominal data:**
Use **distinct colors or shapes** to differentiate categories without implying any order.
Example: Different land use types shown in unique colors.
- **Ordinal data:**
Use **graduated colors or symbol sizes** that reflect the ranked order of data.
Example: Risk levels shown from light to dark color or small to large symbols.
- **Interval and Ratio data:**
Use **proportional symbols** (e.g., varying circle sizes) or **color gradients** to represent differences in magnitude or quantity.
Example: Population density represented by increasingly darker shades or larger circles.

2.4 Pattern Recognition

Pattern recognition in spatial data involves identifying the way features are distributed across an area. Understanding these patterns helps reveal the processes influencing the spatial arrangement.

- **Random:**
Features or points are scattered without any predictable order or pattern.
- **Clustered:**
Features group together in specific areas, showing concentrations or hotspots.
- **Uniform:**
Features are evenly spaced, often due to planned distribution or natural constraints.

2.5 Pattern Analysis and Quantification

Spatial pattern analysis uses statistical methods to measure and quantify the arrangement of features in space. This analysis determines whether features are **clustered**, **dispersed**, or **randomly distributed**.

- **Clustering:** Clustering means how much features are grouped closely together compared to what would happen randomly.
- **Dispersion:** Dispersion means features are spread out evenly, more than what would happen by chance.

- **Randomness:** Shows no specific pattern in feature distribution.

2.6 Result Interpretation and Decision Making

Interpreting spatial patterns helps users understand the underlying processes and relationships in geographic data. This understanding enables informed decision-making in areas such as:

- Urban planning
- Environmental management
- Resource allocation
- Public health
- Disaster response

What is Spatial data?

Spatial data is information that describes the location, shape, and relationships of geographic features on Earth. It includes data about where things are (coordinates), their size, and how they relate to each other in space.